Efficient Estimation of Causal Effects Under Two-Phase Sampling with Error-Prone Outcome and Treatment Measurements

HSPH Biostatistics Colloquium Series

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Roadmap

Background

Methods

Simulation Study

Data Application

Discussion

Motivation

Explosion of the use of electronic health record (EHR) data for conducting observational causal inference studies

- Where, at a high level, one is interested in some measure of causal effect of a treatment A on an outcome of interest Y

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- EHR data is typically cheaper to obtain, fairly representative of patient populations, rich in information on potential confounding factors $\boldsymbol{X}$, and big

But EHR data tends to present numerous challenges

- Including measurement error in outcomes (Y) and treatments (A) of interest

EHR data: in a perfect world

Suppose we aim to estimate the average treatment effect of a binary treatment A on Y

$$\tau = \mathbb{E}[Y(1) - Y(0)]$$

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Y_2	A_2	$\boldsymbol{X}_2$
Y_3	A_3	$\boldsymbol{X}_3$
Y_4	A_4	$\boldsymbol{X}_4$
Y_5	A_5	$\boldsymbol{X}_5$
Y_6	A_6	$\boldsymbol{X}_6$

In a perfect world, we'd have access to the true outcome + treatment values (+ covariates $\boldsymbol{X}$)

EHR data: in ~~a perfect world~~ reality

Y	Y^*	A	A^*	$\boldsymbol{X}$
NA	Y_1^*	NA	A_1^*	$\boldsymbol{X}_1$
NA	Y_2^*	NA	A_2^*	$\boldsymbol{X}_2$
NA	Y_3^*	NA	A_3^*	$\boldsymbol{X}_3$
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In practice, we often only have error-prone measurements of Y and A, denoted Y^* and A^*

- Well-documented that using Y^* and A^* in place of Y and A can lead to severely biased causal effect estimates

EHR data: a study design-based workaround

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
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In practice, can sometimes spend time + money to obtain gold-standard measurements for a random (typically small) subset of this data

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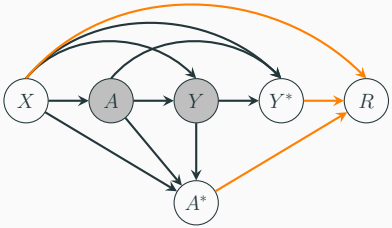

In practice, selection into this subset can often be controlled, and may depend on the initial error-prone measurements: $R \not\perp\!\!\!\perp A^*, Y^*, \boldsymbol{X}$

- Especially common sampling strategy if Y and/or A are rare

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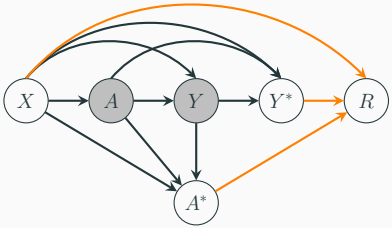

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- **Intuition**: over-sample subjects who contribute more information to the target estimand

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We present two asymptotically equivalent approaches to constructing efficient nonparametric causal effect estimators

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- We document that in general two-phase sampling problems, there are **two separate approaches** to constructing debiased causal effect estimators with desirable properties
  - The resulting estimators are asymptotically equivalent, but their behavior can meaningfully differ in finite samples
  - In practice, researchers commonly take 1 of the 2 approaches, without reference to the other

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- We document that in general two-phase sampling problems, there are **two separate approaches** to constructing debiased causal effect estimators with desirable properties
- Using our measurement error problem as a guiding example, we compare these two approaches
  - We document unique challenges faced by both approaches that can result in erratic finite-sample behavior
  - And present an ensemble estimator which partially protects against these challenges

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- Approach 1
- Approach 2
- Ensemble Estimator

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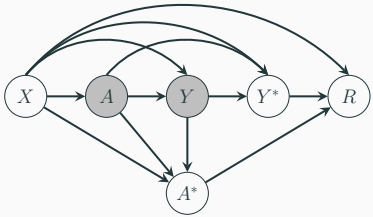

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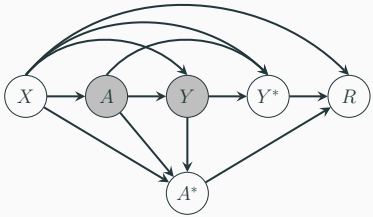

Main goal: Estimate counterfactual means $\mathbb{E}[Y(a)]$ $(a \in \{0,1\})$ *efficiently*, allowing R to depend on $\boldsymbol{X}, Y^*$ and A^*

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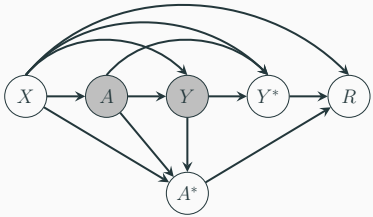

Construction of estimators follows standard semiparametric estimation pipeline:

causal estimand
$\mathbb{E}[Y(a)]$

$\xrightarrow{\text{Assumptions}}$

stat. estimand
$\psi_n = \mathbb{E}\, f(\text{data})$

$\longrightarrow$

plug-in estimator
$\hat{\psi}_n^{\text{PI}} = \frac{1}{n}\sum_{i=1}^{n} f(\text{data}_i)$

$\xrightarrow{\text{(tedious) work}}$

debiased est.
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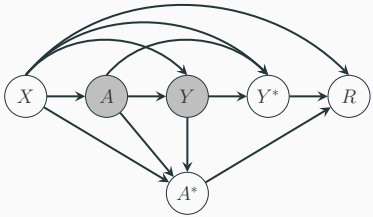

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$$\underset{\text{causal estimand}}{\mathbb{E}[Y(a)]} \xrightarrow[\text{Assumptions}]{} \psi_a = \underset{\text{stat. estimand}}{\mathbb{E}[f(\text{data})]} \longrightarrow \hat{\psi}_a^{\text{Pl}} = \underbrace{\frac{1}{n}\sum_{i=1}^{n} f(\text{data}_i)}_{\text{plug-in estimator}} \xrightarrow[\text{(tedious) work}]{} \underbrace{\hat{\psi}_a^{\text{DR}}}_{\text{debiased est.}}$$

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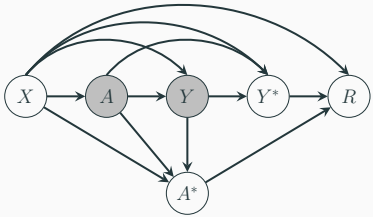

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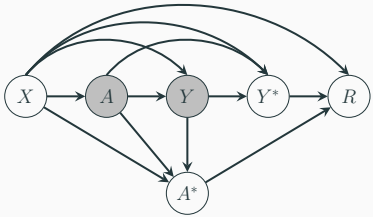

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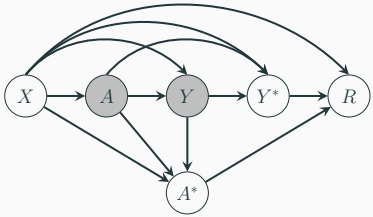

Letting $\boldsymbol{Z} = (\boldsymbol{X}, A^*, Y^*)$, we make the three "core" causal assumptions:

(1) $\underbrace{Y = AY(1) + (1-A)Y(0)}_{\text{Consistency}}$, **(2)** $\underbrace{\mathbb{P}(A = 1|\boldsymbol{X}) \in (0,1)}_{\text{Positivity of treatment}}$, **(3)** $\underbrace{Y(a) \perp\!\!\!\perp A|\boldsymbol{X}}_{\text{Unconfoundedness}}$,

as well as two assumptions related to the missingness mechanism for Y and A:

(4) $\underbrace{\mathbb{P}(R = 1|\boldsymbol{Z}) \in (0,1)}_{\text{Validation positivity}}$, **(5)** $\underbrace{(Y, A) \perp\!\!\!\perp R|\boldsymbol{Z}}_{\text{Missing at random}}$

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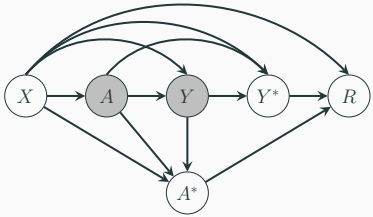

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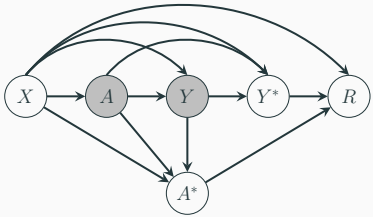

Letting $\boldsymbol{Z} = (\boldsymbol{X}, A^*, Y^*)$, we show:

$$\mathbb{E}[Y(a)] = \psi_a := \mathbb{E}\left[\frac{\mathbb{E}[\lambda_a(\boldsymbol{Z}) \cdot \mu_a(\boldsymbol{Z})|\boldsymbol{X}]}{\mathbb{E}[\lambda_a(\boldsymbol{Z})|\boldsymbol{X}]}\right]$$

where $\lambda_a(\boldsymbol{Z}) := \mathbb{P}(A = a|\boldsymbol{Z}, R = 1)$ and $\mu_a(\boldsymbol{Z}) := \mathbb{E}(Y|A = a, \boldsymbol{Z}, R = 1)$ are <u>imputation functions</u>

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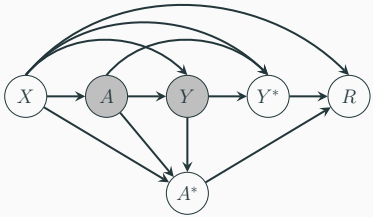

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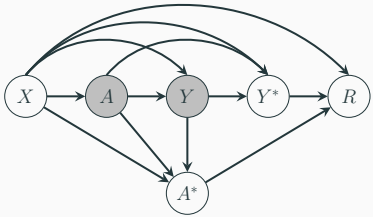

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$$\hat{\psi}_a^{\text{PI,1}} = \frac{1}{n} \sum_{i=1}^{n} \frac{\hat{\mathbb{E}}[\hat{\lambda}_a(\boldsymbol{Z}_i) \cdot \hat{\mu}_a(\boldsymbol{Z}_i) | \boldsymbol{X}_i]}{\hat{\mathbb{E}}[\hat{\lambda}_a(\boldsymbol{Z}_i) | \boldsymbol{X}_i]}$$

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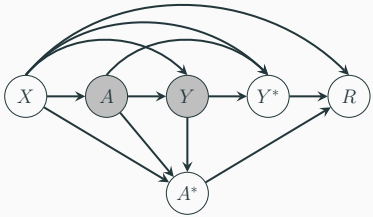

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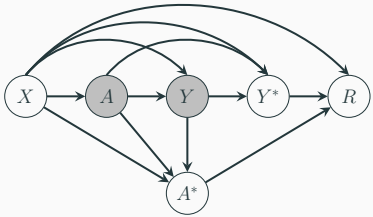
$$\hat{\psi}_a^{\text{PI},1} = \frac{1}{n}\sum_{i=1}^{n} \frac{\hat{\mathbb{E}}[\hat{\lambda}_a(\boldsymbol{Z}_i) \cdot \hat{\mu}_a(\boldsymbol{Z}_i)|\boldsymbol{X}_i]}{\hat{\mathbb{E}}[\hat{\lambda}_a(\boldsymbol{Z}_i)|\boldsymbol{X}_i]}$$

where $\hat{\lambda}_a(\boldsymbol{Z}) = \hat{\mathbb{P}}(A = a|\boldsymbol{Z}, R = 1)$ and $\hat{\mu}_a(\boldsymbol{Z}) = \hat{\mathbb{E}}(Y|A = a, \boldsymbol{Z}, R = 1)$

- Interpretation: IPW on imputed values, after marginalizing out error-prone variables

Approach 1: Using the observed data distribution

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



Letting $\boldsymbol{Z} = (\boldsymbol{X}, A^*, Y^*)$, we construct a plug-in estimator

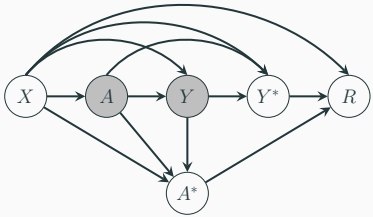
$$\hat{\psi}_a^{\text{PI},1} = \frac{1}{n} \sum_{i=1}^{n} \frac{\hat{\mathbb{E}}[\hat{\lambda}_a(\boldsymbol{Z}_i) \cdot \hat{\mu}_a(\boldsymbol{Z}_i) | \boldsymbol{X}_i]}{\hat{\mathbb{E}}[\hat{\lambda}_a(\boldsymbol{Z}_i) | \boldsymbol{X}_i]}$$

where $\hat{\lambda}_a(\boldsymbol{Z}) = \hat{\mathbb{P}}(A = a | \boldsymbol{Z}, R = 1)$ and $\hat{\mu}_a(\boldsymbol{Z}) = \hat{\mathbb{E}}(Y | A = a, \boldsymbol{Z}, R = 1)$

- Drawback: Inference intractable when nuisance models are fit data-adaptively

Approach 1: Using the observed data distribution

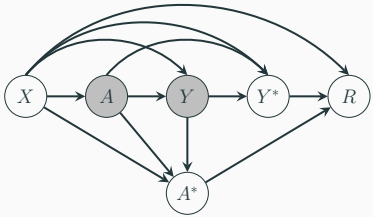
Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



To enable inference, we derive a one-step debiased estimator

Approach 1: Using the observed data distribution

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



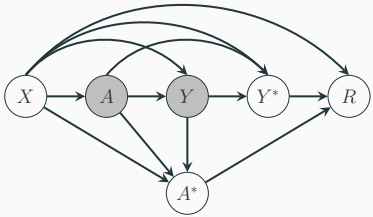
To enable inference, we derive a one-step debiased estimator

$$\hat{\psi}_a^{\text{OS},1} = \frac{1}{n}\sum_{i=1}^{n} \frac{\hat{\mathbb{E}}[\hat{\lambda}_a(\boldsymbol{Z}_i)\cdot\hat{\mu}_a(\boldsymbol{Z}_i)|\boldsymbol{X}_i]}{\hat{\mathbb{E}}[\hat{\lambda}_a(\boldsymbol{Z}_i)|\boldsymbol{X}_i]} + \widehat{\text{BC}}$$

where $\widehat{\text{BC}}$ is a bias correction term based on the efficient influence function for the statistical estimand, ψ_a

Approach 1: Using the observed data distribution

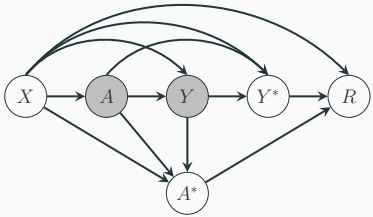
Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



We document multiple properties of $\hat{\psi}_a^{\text{OS},1}$

Approach 1: Using the observed data distribution

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1

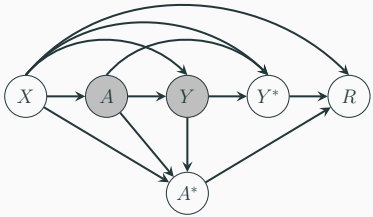


We document multiple properties of $\hat{\psi}_a^{\text{OS},1}$

1. Bias correction enables valid inference when nuisance models are fit with flexible ML methods that converge at $n^{1/4}$ rates, however...

Approach 1: Using the observed data distribution

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



We document multiple properties of $\hat{\psi}_a^{\text{OS},1}$

1. Bias correction enables valid inference when nuisance models are fit with flexible ML methods that converge at $n^{1/4}$ rates, however...

2. This bias correction term introduces numerous unstable weighting terms that can harm finite sample performance View weighting terms
   - Particularly concerning, as validation samples tend to be small in practice

Roadmap

Background

Methods

- Approach 1
- Approach 2
- Ensemble Estimator

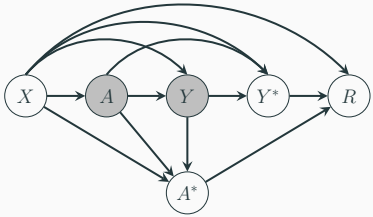
Simulation Study

Data Application

Discussion

Approach 2: Complete-data projection

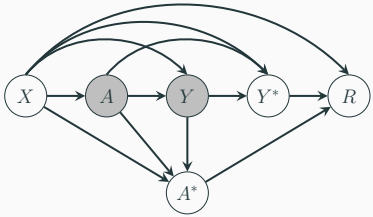
Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



Approach 2 is based on a well-developed, but relatively underutilized, framework for constructing debiased estimators under missing data (van der Laan and Robins 2003)

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1

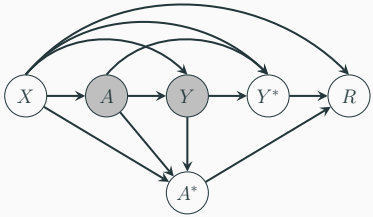


Approach 2 is based on a well-developed, but relatively underutilized, framework for constructing debiased estimators under missing data (van der Laan and Robins 2003)

- Links (i) the estimator we'd ideally construct under complete data to (ii) the observed data structure, where Y and A are partially missing

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
Y_5	Y_5^*	A_5	A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



Idea: If we had complete data, could construct a plug-in estimator $\hat{\psi}_a^{\mathsf{PI},2} = \frac{1}{n}\sum_{i=1}^n \hat{m}_a(\boldsymbol{X}_i)$,

$$\hat{\psi}_a^{\mathsf{OS},1} = \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^n \left(m_a(\boldsymbol{X}_i) + \frac{I\{A_i = a\}}{g_a(\boldsymbol{X}_i)}\{Y_i - m_a(\boldsymbol{X}_i)\} - \hat{\psi}_a^{\mathsf{PI},2} \right)$$

where $m_a(\boldsymbol{X}) = \mathbb{E}(Y|A = a, \boldsymbol{X})$ and $g_a(\boldsymbol{X}) = \mathbb{P}(A = a|\boldsymbol{X})$

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
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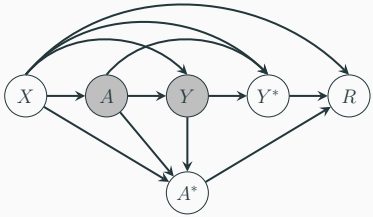
Y_6	Y_6^*	A_6	A_6^*	X_6	1



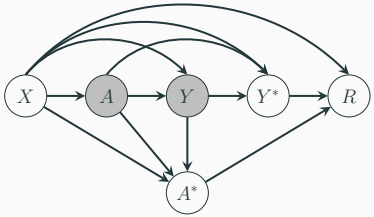
Idea: If we had complete data, could construct a plug-in estimator $\hat{\psi}_a^{\mathsf{PI},2} = \frac{1}{n}\sum_{i=1}^n \hat{m}_a(\boldsymbol{X}_i)$, as well as an augmented inverse probability weighted (AIPW) estimator

$$\hat{\psi}_a^{\mathsf{OS},2} = \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^n \left(\hat{m}_a(\boldsymbol{X}_i) + \frac{I(A_i = a)}{\hat{g}_a(\boldsymbol{X}_i)}\{Y_i - \hat{m}_a(\boldsymbol{X}_i)\} - \hat{\psi}_a^{\mathsf{PI},2} \right)$$

where $m_a(\boldsymbol{X}) = \mathbb{E}(Y|A = a, \boldsymbol{X})$ and $g_a(\boldsymbol{X}) = \mathbb{P}(A = a|\boldsymbol{X})$

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
Y_5	Y_5^*	A_5	A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



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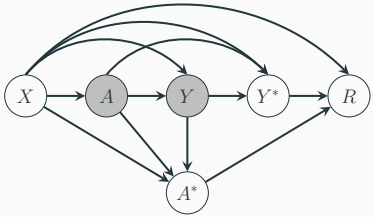
$$\hat{\psi}_a^{\text{OS},2} = \hat{\psi}_a^{\text{PI},2} + \frac{1}{n}\sum_{i=1}^n \left(\hat{m}_a(\boldsymbol{X}_i) + \frac{I(A_i = a)}{\hat{g}_a(\boldsymbol{X}_i)}\{Y_i - \hat{m}_a(\boldsymbol{X}_i)\} - \hat{\psi}_a^{\text{PI},2} \right)$$

Above, $m_a(\boldsymbol{X})$ and $g_a(\boldsymbol{X})$ can be estimated with <u>weighted regressions</u> that add weights $R/\hat{\mathbb{P}}(R = 1|\boldsymbol{Z})$ to the underlying loss function (Rose and van der Laan, 2011)

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
Y_5	Y_5^*	A_5	A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



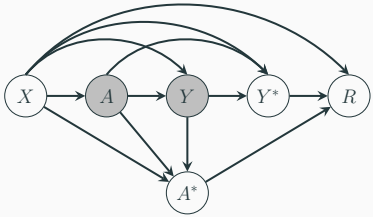
$$\hat{\psi}_a^{\mathsf{OS},2} = \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^{n}\left(\hat{m}_a(\boldsymbol{X}_i) + \frac{I(A_i = a)}{\hat{g}_a(\boldsymbol{X}_i)}\{Y_i - \hat{m}_a(\boldsymbol{X}_i)\} - \hat{\psi}_a^{\mathsf{PI},2}\right)$$

Issue: The bias correction terms are <u>only observed</u> when $R_i = 1$

- Above estimator is infeasible

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
Y_5	Y_5^*	A_5	A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



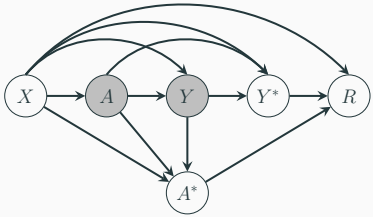
Key idea: Treat the bias correction terms as pseudo-outcomes:

$$\hat{\psi}_a^{\mathsf{OS},2} = \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^{n}\left(\hat{m}_a(\boldsymbol{X}_i) + \frac{I(A_i = a)}{\hat{g}_a(\boldsymbol{X}_i)}\{Y_i - \hat{m}_a(\boldsymbol{X}_i)\} - \hat{\psi}_a^{\mathsf{PI},2}\right)$$

$$= \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^{n}\underbrace{\chi_a(\boldsymbol{O}_i; \hat{m}_a, \hat{g}_a)}_{\text{pseudo outcome}}$$

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
Y_5	Y_5^*	A_5	A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1

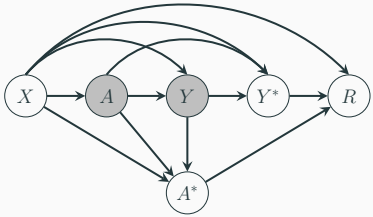


Key idea: Perform AIPW on the pseudo-outcome:

$$\hat{\psi}_a^{\mathsf{OS},2} = \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^{n} \chi_a(\boldsymbol{O}_i; \hat{m}_a, \hat{g}_a)$$

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
Y_5	Y_5^*	A_5	A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



Key idea: Perform AIPW on the pseudo-outcome:

$$\hat{\psi}_a^{\mathsf{OS},2} = \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^{n}\left(\hat{\varphi}_a(\boldsymbol{Z}_i) + \frac{R_i}{\mathbb{P}(R_i = 1|\boldsymbol{Z}_i)}\{\chi_a(\boldsymbol{O}_i; \hat{m}_a, \hat{g}_a) - \hat{\varphi}_a(\boldsymbol{Z}_i)\}\right)$$

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
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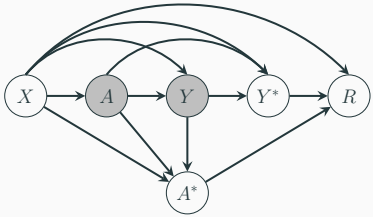


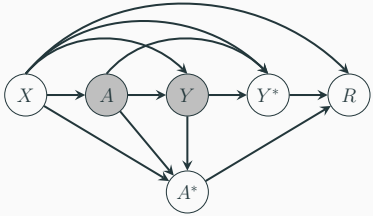
Key idea: Perform AIPW on the pseudo-outcome:

$$\hat{\psi}_a^{\text{OS},2} = \hat{\psi}_a^{\text{PI},2} + \frac{1}{n}\sum_{i=1}^{n}\left(\hat{\varphi}_a(\boldsymbol{Z}_i) + \frac{R_i}{\mathbb{P}(R_i = 1|\boldsymbol{Z}_i)}\{\chi_a(\boldsymbol{O}_i; \hat{m}_a, \hat{g}_a) - \hat{\varphi}_a(\boldsymbol{Z}_i)\}\right)$$

where $\hat{\varphi}_a(\boldsymbol{Z}) = \hat{\mathbb{E}}[\chi_a(\boldsymbol{O}; \hat{m}_a, \hat{g}_a)|\boldsymbol{Z}, R = 1]$

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
Y_5	Y_5^*	A_5	A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



Key idea: Perform AIPW on the pseudo-outcome:

$$\hat{\psi}_a^{\mathsf{OS},2} = \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^{n}\left(\hat{\varphi}_a(\boldsymbol{Z}_i) + \frac{R_i}{\mathbb{P}(R_i = 1|\boldsymbol{Z}_i)}\{\chi_a(\boldsymbol{O}_i; \hat{m}_a, \hat{g}_a) - \hat{\varphi}_a(\boldsymbol{Z}_i)\}\right)$$

where $\hat{\varphi}_a(\boldsymbol{Z}) = \hat{\mathbb{E}}[\chi_a(\boldsymbol{O}_i; \hat{m}_a, \hat{g}_a)|\boldsymbol{Z}, R = 1]$

- Feasible estimator!
- When sampling probabilities are known, $\hat{\psi}_a^{\mathsf{OS},2}$ is doubly-robust in the traditional sense

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
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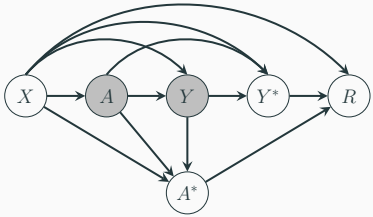
Y_6	Y_6^*	A_6	A_6^*	X_6	1



Specifically, letting $\kappa(\boldsymbol{Z}) := \mathbb{P}(R = 1|\boldsymbol{Z})$ we show

$$\mathbb{E}[\hat{\psi}_a^{\mathsf{OS},2} - \psi_a] = O_{\mathbb{P}}\left(\frac{1}{\sqrt{n}} \quad + \quad ||\hat{m}_a - m_a|| \cdot ||\hat{g}_a - g_a|| \quad + \quad ||\hat{\kappa} - \kappa|| \cdot ||\hat{\varphi}_a - \varphi_a||\right)$$

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
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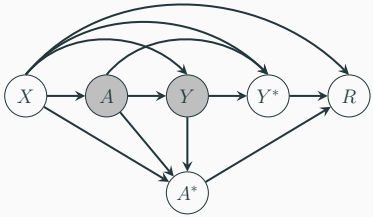

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$$\mathbb{E}[\hat{\psi}_a^{\mathsf{OS},2} - \psi_a] = O_{\mathbb{P}}\Bigg(\frac{1}{\sqrt{n}} \quad + \quad \underbrace{||\hat{m}_a - m_a|| \cdot ||\hat{g}_a - g_a||}_{\text{complete data}} \quad + \quad \underbrace{||\hat{\kappa} - \kappa|| \cdot ||\hat{\varphi}_a - \varphi_a||}_{\text{missingness mechanism}}\Bigg)$$

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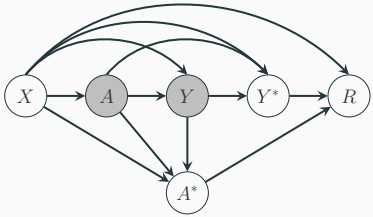

Connections between $\hat{\psi}_a^{\text{OS},1}$ and $\hat{\psi}_a^{\text{OS},2}$:

- Can be shown that asymptotically, Approach 1 and Approach 2 estimators are *equivalent*
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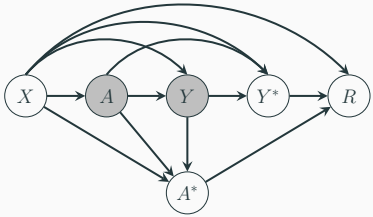

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  - Approach 1: Bias correction introduces unstable weighting terms

Approach 2: Complete-data projection

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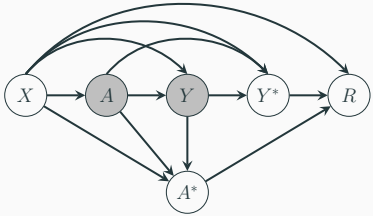

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  - Approach 2: The pseudo-outcome regression function $\varphi_a(\boldsymbol{Z})$ can be difficult to estimate in small samples

Roadmap

Background

Methods

- Approach 1
- Approach 2
- Ensemble Estimator

Simulation Study

Data Application

Discussion

Approaches 1 and 2

Previous slides suggest

- $\hat{\psi}_a^{\text{OS},1}$ and $\hat{\psi}_a^{\text{OS},2}$ have the same asymptotic distribution, but...
- In finite samples, their behavior can meaningfully differ due to unique sources of instability

Finite sample performance is crucial in two-phase settings, since phase-two samples will often be small

Challenge: Choosing to report an estimator based on post-hoc efficiency checks invalidates inference

Ensemble Estimator

Can avoid the need to choose an estimator *a priori* by taking a variance-optimal weighted average of the two one-step estimators

$$\hat{\psi}_a^{\mathsf{OS,W}} = \hat{w} \cdot \hat{\psi}_a^{\mathsf{OS,1}} + (1 - \hat{w}) \cdot \hat{\psi}_a^{\mathsf{OS,2}}$$

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Advantage: Provides a means to pre-register an estimator of $\mathbb{E}[Y(a)]$ that is agnostic to whether $\hat{\psi}_a^{\mathsf{OS,1}}$ or $\hat{\psi}_a^{\mathsf{OS,2}}$ will be more 'efficient

Note: many variations of the above approach

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Note: many variations of the above approach

- Similar approaches to the one above have been considered in data fusion (Karlsson et al., 2024) and high-dimensional estimation (Antonelli and Cefalu, 2020) settings

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Methods

Simulation Study

Data Application

Discussion

Simulation: Setup

Full details

We simulate data according to a two-phase sampling procedure dependent on A^*, Y^* and $\boldsymbol{X}$, varying **(i)** the overall sample size, n, and **(ii)** the share of validation data $\mathbb{P}(R = 1)$

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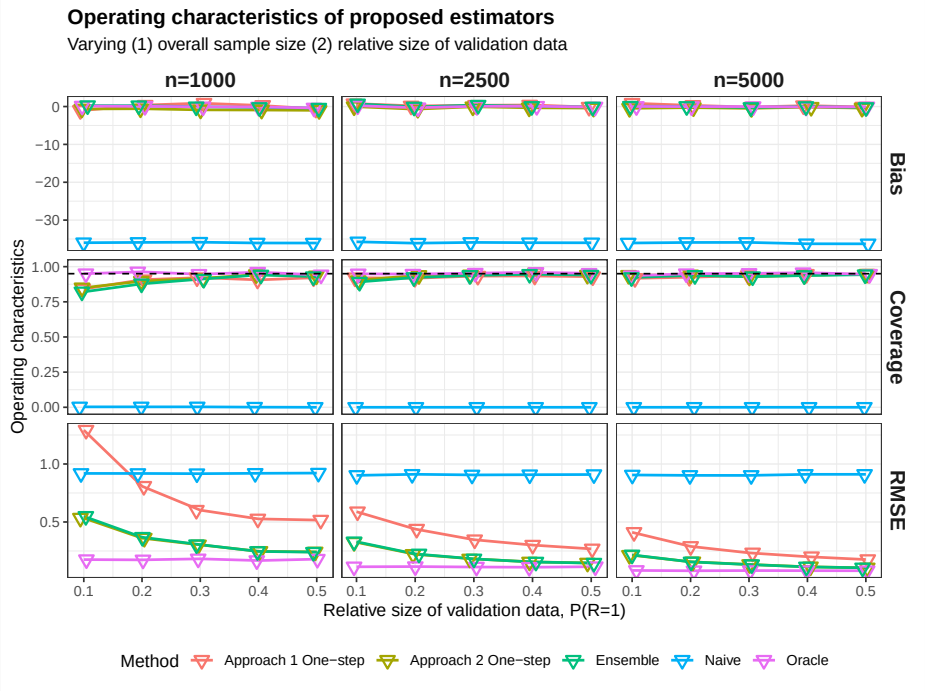
Estimators:

- An oracle AIPW estimator which has access to the true A and Y for all units, and a naive AIPW estimator which uses A^* and Y^* for all units
- The proposed Approach 1 and 2 one-step estimators
- The ensemble estimator $\hat{\psi}^{\text{OS,W}}$

All nuisance functions are fit with a stacked ensemble (Super Learner) of GLMs and GAMs


Operating characteristics of proposed estimators
Varying (1) overall sample size (2) relative size of validation data
n=1000
n=2500
n=5000
Bias
Coverage
RMSE
Operating characteristics
0
−10
−20
−30
1.00
0.75
0.50
0.25
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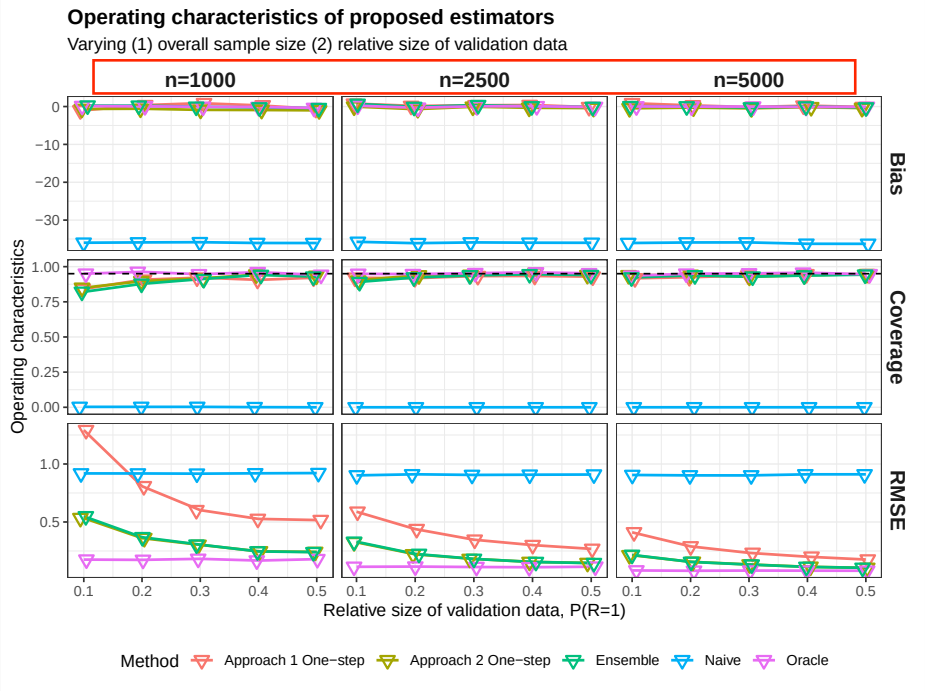
Relative size of validation data, P(R=1)
Method
Approach 1 One−step
Approach 2 One−step
Ensemble
Naive
Oracle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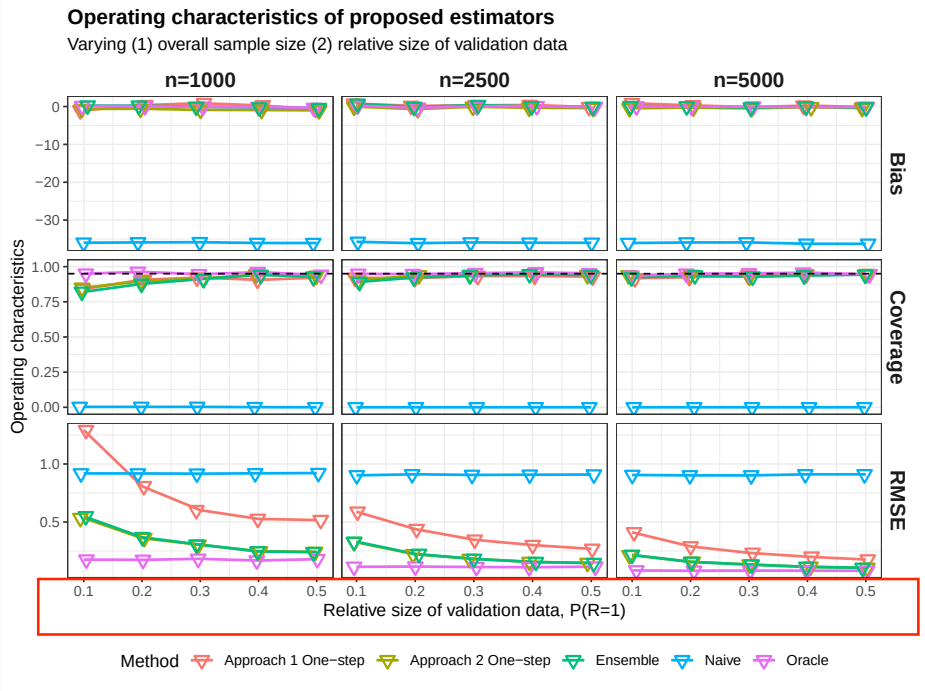
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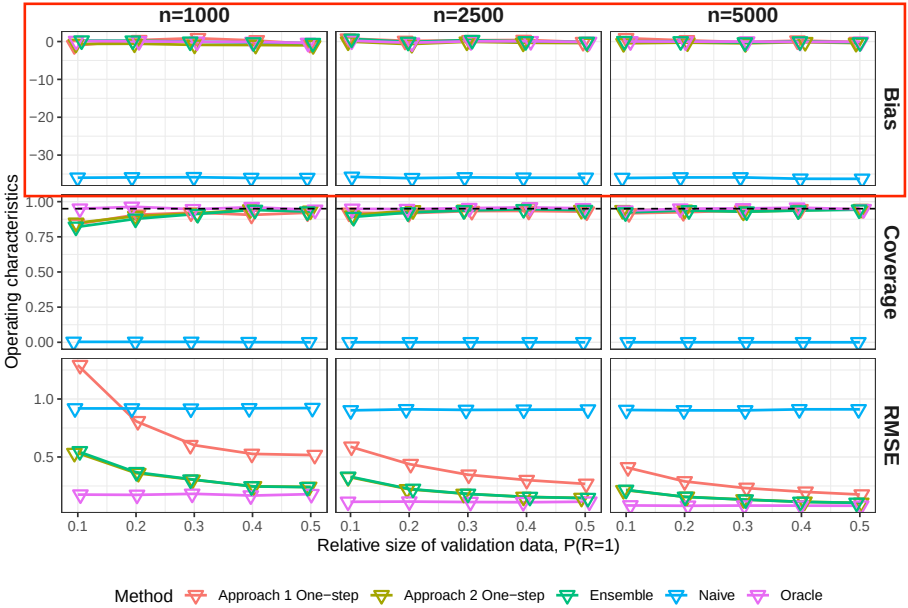

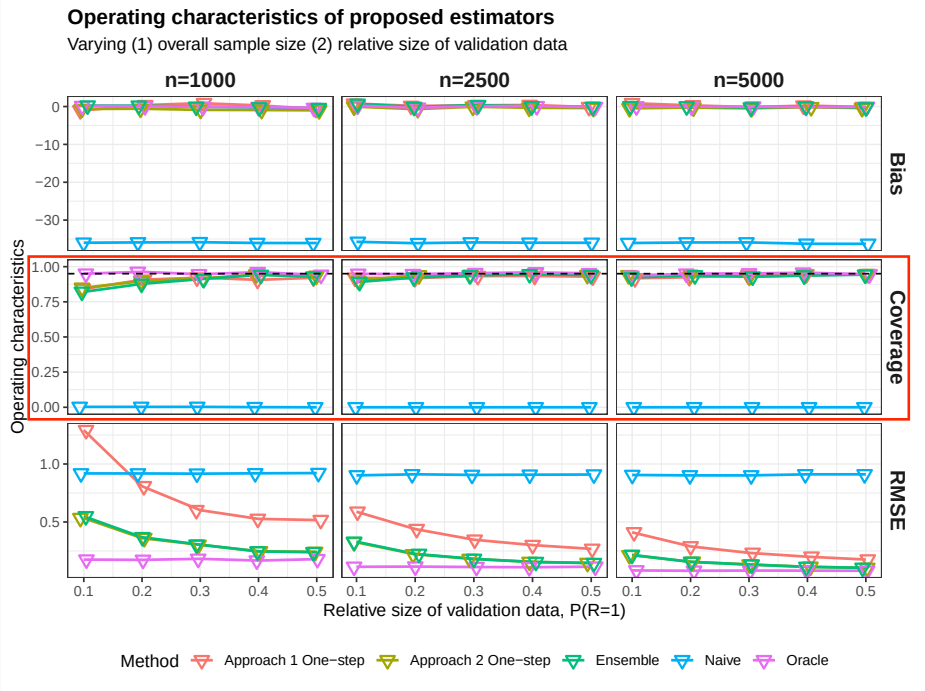
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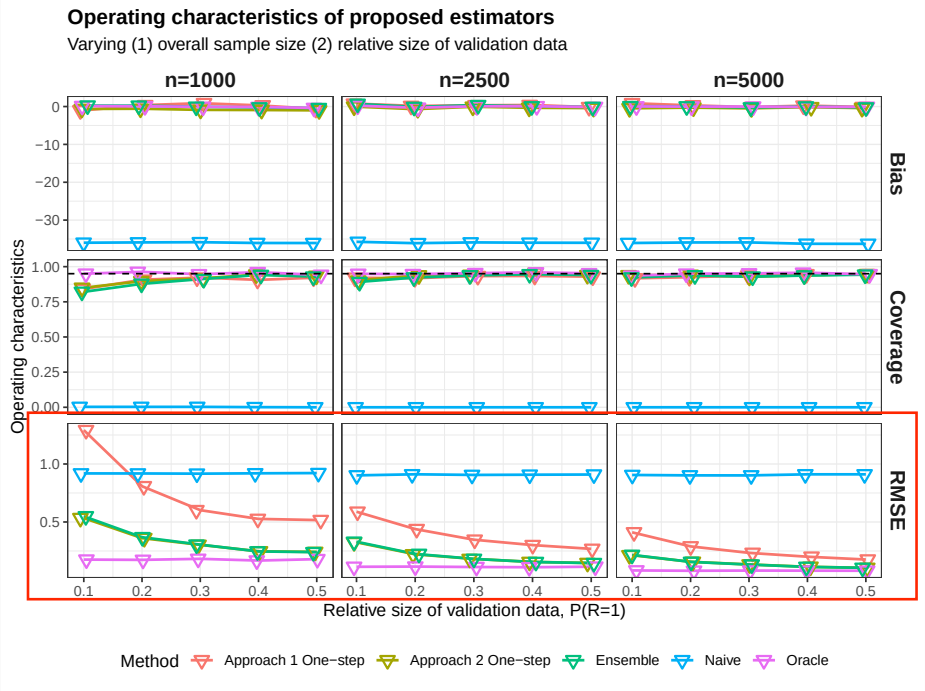
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Data Application

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EHR database with information on $\approx$ 1900 patients living with HIV between 1998-2011 that began receiving care at the VCCC

- Numerous variables recorded with error
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Causal estimand: Average causal effect of starting ART within 1 month of first visit (A) on 3-year (post initial visit) risk of suffering an ADE (Y)

VCCC application

"Reveal" increasingly larger shares of the validated data $\mathbb{P}(R = 1)$, over-sampling those with $A^* = 1$ and $Y^* = 1$

- E.g. 5%, 10%, $15\%, \ldots$
- Implement proposed estimators at each share

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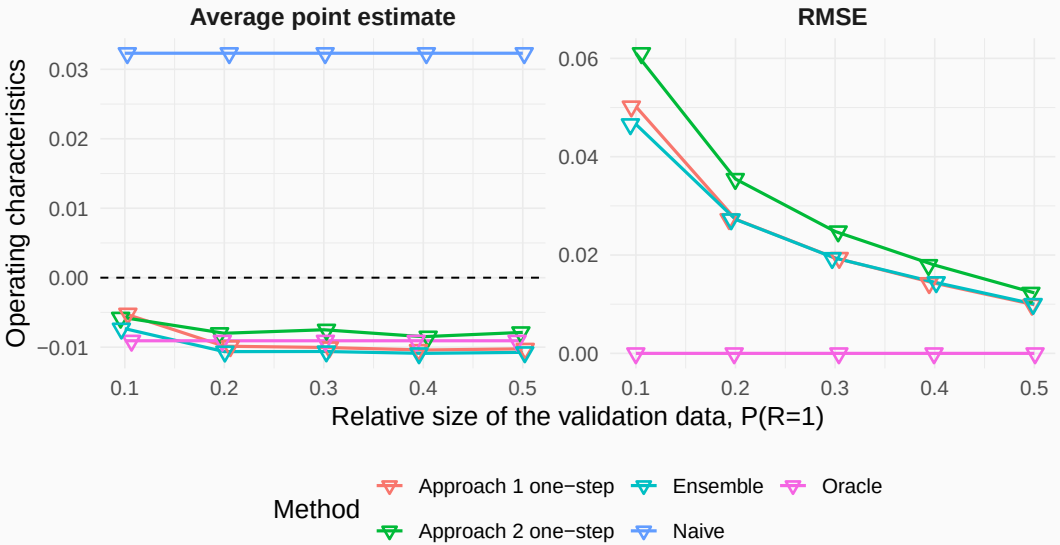
Operating characteristics: 3−year ADE risk

Treatment: ART initiation within one month of first visit



Roadmap

Background

Methods

Simulation Study

Data Application

Discussion

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Growth in use of information-dependent two-phase sampling schemes, particularly in the presence of joint outcome and treatment measurement error

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Growth in use of information-dependent two-phase sampling schemes, particularly in the presence of joint outcome and treatment measurement error

We've presented debiased one-step estimators that accommodate these sampling schemes

- Ensemble estimator will tend to be as or more efficient than both $\hat{\psi}_a^{\mathsf{OS},1}$ and $\hat{\psi}_a^{\mathsf{OS},2}$ in finite samples
- We developed the R package `drcmd`, which implements Approach 2 for general missing data patterns

Notably, Approaches 1 and 2 are applicable in general two-phase sampling / missing data problems

Future Directions

Adaptive determination of two-phase sampling probabilities $\mathbb{P}(R = 1|\boldsymbol{Z})$

- while allowing for non-monotone missingness patterns and variable-dependent two-phase sampling costs

Exploration of the ensemble approach in broader estimation problems

- e.g. can ensemble AIPW and TMLE estimators

Automatic stabilization methods for nonlinear functionals

- Recalling the Approach 1 method targeted a nonlinear functional of the data
- Exploring analogues of Riesz regression Chernozhukov et al. (2018)

Thank you!

https://arxiv.org/pdf/2506.21777

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Appendix

Identification for Approach 1

Back to main slide

First, let $\boldsymbol{W} = (Y^*, A^*)$. Then, we have

$$\begin{aligned}
\mathbb{E}(Y(a)) &= \mathbb{E}_{\boldsymbol{X}}\mathbb{E}(Y(a)|\boldsymbol{X}) \\
&= \mathbb{E}_{\boldsymbol{W}(a)|\boldsymbol{X}}\ [\mathbb{E}_{\boldsymbol{X}}\mathbb{E}(Y(a)|\boldsymbol{X}, \boldsymbol{W}(a))] \\
&= \mathbb{E}_{\boldsymbol{W}(a)|\boldsymbol{X}}\ [\mathbb{E}_{\boldsymbol{X}}\mathbb{E}(Y(a)|\boldsymbol{X}, \boldsymbol{W}(a), A = a, R = 1)] \\
&= \mathbb{E}_{\boldsymbol{W}|\boldsymbol{X},A}\ [\mathbb{E}_{\boldsymbol{X}}\mathbb{E}(Y|\boldsymbol{X}, \boldsymbol{W}, A = a, R = 1)]
\end{aligned}$$

The problem: the density $p(w|x, a)$ is unidentified

- We only see A in the validation data, which is **not** independent of $\boldsymbol{W}$ so we can't just condition on $R = 1$

Identification for Approach 1

Notice

$$\mathbb{E}(Y(a)) = \sum_{w} \sum_{x} \sum_{y} y \cdot p(y|w, x, a, r = 1) \cdot p(w|x, a) \cdot p(x)$$

Identification for Approach 1

Notice

$$\mathbb{E}(Y(a)) = \sum_w \sum_x \sum_y y \cdot p(y|w,x,a,r=1) \cdot p(w|x,a) \cdot p(x)$$

$$= \sum_w \sum_x \sum_y y \cdot p(y|w,x,a,r=1) \cdot \frac{p(a|x,w)p(w|x)p(x)}{p(a|x)p(x)} \cdot p(x)$$

Identification for Approach 1

Notice

$$
\begin{aligned}
\mathbb{E}(Y(a)) &= \sum_w \sum_x \sum_y y \cdot p(y|w,x,a,r=1) \cdot p(w|x,a) \cdot p(x) \\
&= \sum_w \sum_x \sum_y y \cdot p(y|w,x,a,r=1) \cdot \frac{p(a|x,w)p(w|x)p(x)}{p(a|x)p(x)} \cdot p(x) \\
&= \sum_w \sum_x \sum_y y \cdot p(y|w,x,a,r=1) \cdot \frac{p(a|x,w,r=1)p(w|x)}{\sum_{w'} p(a|w',x)p(w'|x)} \cdot p(x)
\end{aligned}
$$

Identification for Approach 1

Notice

$$
\begin{aligned}
\mathbb{E}(Y(a)) &= \sum_w \sum_x \sum_y y \cdot p(y|w,x,a,r=1) \cdot p(w|x,a) \cdot p(x) \\
&= \sum_w \sum_x \sum_y y \cdot p(y|w,x,a,r=1) \cdot \frac{p(a|x,w)p(w|x)p(x)}{p(a|x)p(x)} \cdot p(x) \\
&= \sum_w \sum_x \sum_y y \cdot p(y|w,x,a,r=1) \cdot \frac{p(a|x,w,r=1)p(w|x)}{\sum_{w'} p(a|w',x)p(w'|x)} \cdot p(x)
\end{aligned}
$$

Key idea: Use Bayes rule to re-express un-identified density in terms of identifiable ones

- Then, overall expression is identified – just need to define terms and "collapse" the above expectations

Identification for Approach 1

Recall that $\lambda_a(\boldsymbol{Z}) = \mathbb{P}(A = a|\boldsymbol{Z}, R = 1)$ and $\mu_a(\boldsymbol{Z}) = \mathbb{E}[Y|A = a, \boldsymbol{Z}, R = 1]$. Previous slide implies

$$\mathbb{E}[Y(a)] = \mathbb{E}\left[\frac{\mathbb{E}[\lambda_a(\boldsymbol{Z}) \cdot \mu_a(\boldsymbol{Z})|\boldsymbol{X}]}{\mathbb{E}[\lambda_a(\boldsymbol{Z})|\boldsymbol{X}]}\right]$$

Plug-in implementation

Y_i	Y_i^*	A_i	A_i^*	$\boldsymbol{X}_i$	R_i	$\hat{\lambda}_a$	$\hat{\mu}_a$	$\hat{\pi}_a$	$\hat{\eta}_a$
1	0	0	1	3.2	1	NA	NA	NA	NA
0	1	1	1	2.1	1	NA	NA	NA	NA
1	1	0	0	0.3	1	NA	NA	NA	NA
1	0	0	1	1.6	0	NA	NA	NA	NA
0	1	0	1	4.8	0	NA	NA	NA	NA
1	1	0	0	0.9	0	NA	NA	NA	NA

Step 1: use validation data to fit $\hat{\lambda}_a$ and $\hat{\mu}_a$

Plug-in implementation

Y_i	Y_i^*	A_i	A_i^*	$\boldsymbol{X}_i$	R_i	$\hat{\lambda}_a$	$\hat{\mu}_a$	$\hat{\pi}_a$	$\hat{\eta}_a$
1	0	0	1	3.2	1	0.2	0.72	NA	NA
0	1	1	1	2.1	1	0.43	0.18	NA	NA
1	1	0	0	0.3	1	0.64	0.77	NA	NA
1	0	0	1	1.6	0	0.03	0.55	NA	NA
0	1	0	1	4.8	0	0.81	0.11	NA	NA
1	1	0	0	0.9	0	0.39	0.83	NA	NA

Step 1: use validation data to fit $\hat{\lambda}_a$ and $\hat{\mu}_a$

Plug-in implementation

Y_i	Y_i^*	A_i	A_i^*	$\boldsymbol{X}_i$	R_i	$\hat{\lambda}_a$	$\hat{\mu}_a$	$\hat{\pi}_a$	$\hat{\eta}_a$
1	0	0	1	3.2	1	0.2	0.72	NA	NA
0	1	1	1	2.1	1	0.43	0.18	NA	NA
1	1	0	0	0.3	1	0.64	0.77	NA	NA
1	0	0	1	1.6	0	0.03	0.55	NA	NA
0	1	0	1	4.8	0	0.81	0.11	NA	NA
1	1	0	0	0.9	0	0.39	0.83	NA	NA

Step 2: regress $\hat{\lambda}_a$ and $\hat{\mu}_a$ on $\boldsymbol{Z}$ to yield $\hat{\pi}_a$ and $\hat{\eta}_a$

Plug-in implementation

Y_i	Y_i^*	A_i	A_i^*	$\boldsymbol{X}_i$	R_i	$\hat{\lambda}_a$	$\hat{\mu}_a$	$\hat{\pi}_a$	$\hat{\eta}_a$
1	0	0	1	3.2	1	0.2	0.72	0.77	0.23
0	1	1	1	2.1	1	0.43	0.18	0.41	0.26
1	1	0	0	0.3	1	0.64	0.77	0.59	0.80
1	0	0	1	1.6	0	0.03	0.55	0.08	0.53
0	1	0	1	4.8	0	0.81	0.11	0.72	0.18
1	1	0	0	0.9	0	0.39	0.83	0.38	0.79

Step 2: regress $\hat{\lambda}_a$ and $\hat{\mu}_a$ on $\boldsymbol{Z}$ to yield $\hat{\pi}_a$ and $\hat{\eta}_a$

Plug-in implementation

Y_i	Y_i^*	A_i	A_i^*	$\boldsymbol{X}_i$	R_i	$\hat{\lambda}_a$	$\hat{\mu}_a$	$\hat{\pi}_a$	$\hat{\eta}_a$
1	0	0	1	3.2	1	0.2	0.72	0.77	0.23
0	1	1	1	2.1	1	0.43	0.18	0.41	0.26
1	1	0	0	0.3	1	0.64	0.77	0.59	0.80
1	0	0	1	1.6	0	0.03	0.55	0.08	0.53
0	1	0	1	4.8	0	0.81	0.11	0.72	0.18
1	1	0	0	0.9	0	0.39	0.83	0.38	0.79

Step 3: construct $\hat{\psi}_a^{\mathsf{PI}} = \frac{1}{n}\sum_{i=1}^{n} \frac{\hat{\eta}_a(\boldsymbol{X}_i)}{\hat{\pi}_a(\boldsymbol{X}_i)}$

Simulation Details

Back to main slide

$$\boldsymbol{X} = (X_1, X_2, X_3) \sim \text{Uniform}(0,1) \qquad \text{(Covariates)}$$

$$A|\boldsymbol{X} \sim \text{Bernoulli}(\ \text{expit}(\boldsymbol{X}\boldsymbol{\delta})\ ) \qquad \text{(Treatment)}$$

$$A^* = A \text{ w.p. } 0.8, \quad 1 - A \text{ w.p. } 0.2 \qquad \text{(Treatment measurement)}$$

$$Y = \boldsymbol{X}\boldsymbol{\beta} + \tau A + A\boldsymbol{X}\boldsymbol{\gamma} + \varepsilon, \quad \varepsilon \sim N(0, X_1 + X_2 + X_3) \qquad \text{(Outcome)}$$

$$Y^* = Y + \boldsymbol{X}\boldsymbol{\nu} + v, \quad v \sim N(0,1) \qquad \text{(Outcome measurement)}$$

$$R|\boldsymbol{X}, A^*, Y^* \sim \text{Bernoulli}\left(\rho \times \frac{\text{expit}(\boldsymbol{Z}\boldsymbol{\theta})}{\mathbb{E}[\text{expit}(\boldsymbol{Z}\boldsymbol{\theta})]}\right) \qquad \text{(Validation sampling)}$$

Approach 1: Stability issues

Back to main slide

Recall that $\lambda_a(\boldsymbol{Z}) = \mathbb{P}(A = a|\boldsymbol{Z}, R = 1)$ and $\mu_a(\boldsymbol{Z}) = \mathbb{E}[Y|A = a, \boldsymbol{Z}, R = 1]$ and define

$$\pi_a(\boldsymbol{X}) = \mathbb{E}[\lambda_a(\boldsymbol{Z})|\boldsymbol{X}] \quad \text{and} \quad \eta_a(\boldsymbol{X}) = \mathbb{E}[\lambda_a(\boldsymbol{Z}) \cdot \mu_a(\boldsymbol{Z})|\boldsymbol{X}]$$

Then

$$\begin{aligned}
\hat{\psi}_a^{\mathsf{OS},1} = \frac{1}{n}\sum_{i=1}^{n} \Bigg\{ & \frac{\hat{\eta}_a(\boldsymbol{X}_i)}{\hat{\pi}_a(\boldsymbol{X}_i)} \\
& + \frac{\hat{\lambda}_a(\boldsymbol{Z}_i)}{\hat{\pi}_a(\boldsymbol{X}_i)} \left(\hat{\mu}_a(\boldsymbol{Z}_i) - \frac{\hat{\eta}_a(\boldsymbol{X}_i)}{\hat{\pi}_a(\boldsymbol{X}_i)} \right) \\
& + \frac{R_i I(A_i = a)}{\hat{\mathbb{P}}(R_i = 1|\boldsymbol{Z}_i)\hat{\pi}_a(\boldsymbol{X}_i)} \left(Y_i - \frac{\hat{\eta}_a(\boldsymbol{X}_i)}{\hat{\pi}_a(\boldsymbol{X}_i)} \right) \\
& - \frac{R_i \hat{\lambda}_a(\boldsymbol{Z}_i)}{\hat{\mathbb{P}}(R_i = 1|\boldsymbol{Z}_i)\hat{\pi}_a(\boldsymbol{X}_i)} \left(\hat{\mu}_a(\boldsymbol{Z}_i) - \frac{\hat{\eta}_a(\boldsymbol{X}_i)}{\hat{\pi}_a(\boldsymbol{X}_i)} \right) \Bigg\}
\end{aligned}$$

Approach 1: Stability issues

Back to main slide

Recall that $\lambda_a(\boldsymbol{Z}) = \mathbb{P}(A = a|\boldsymbol{Z}, R = 1)$ and $\mu_a(\boldsymbol{Z}) = \mathbb{E}[Y|A = a, \boldsymbol{Z}, R = 1]$ and define

$$\pi_a(\boldsymbol{X}) = \mathbb{E}[\lambda_a(\boldsymbol{Z})|\boldsymbol{X}] \quad \text{and} \quad \eta_a(\boldsymbol{X}) = \mathbb{E}[\lambda_a(\boldsymbol{Z}) \cdot \mu_a(\boldsymbol{Z})|\boldsymbol{X}]$$

Then

$$\begin{aligned}
\hat{\psi}_a^{\mathsf{OS},1} = \frac{1}{n}\sum_{i=1}^{n} \Bigg\{ & \frac{\hat{\eta}_a(\boldsymbol{X}_i)}{\hat{\pi}_a(\boldsymbol{X}_i)} \\
& + \frac{\hat{\lambda}_a(\boldsymbol{Z}_i)}{\hat{\pi}_a(\boldsymbol{X}_i)} \left(\hat{\mu}_a(\boldsymbol{Z}_i) - \frac{\hat{\eta}_a(\boldsymbol{X}_i)}{\hat{\pi}_a(\boldsymbol{X}_i)} \right) \\
& + \frac{R_i I(A_i = a)}{\hat{\mathbb{P}}(R_i = 1|\boldsymbol{Z}_i)\hat{\pi}_a(\boldsymbol{X}_i)} \left(Y_i - \frac{\hat{\eta}_a(\boldsymbol{X}_i)}{\hat{\pi}_a(\boldsymbol{X}_i)} \right) \\
& - \frac{R_i \hat{\lambda}_a(\boldsymbol{Z}_i)}{\hat{\mathbb{P}}(R_i = 1|\boldsymbol{Z}_i)\hat{\pi}_a(\boldsymbol{X}_i)} \left(\hat{\mu}_a(\boldsymbol{Z}_i) - \frac{\hat{\eta}_a(\boldsymbol{X}_i)}{\hat{\pi}_a(\boldsymbol{X}_i)} \right) \Bigg\}
\end{aligned}$$

Subjects with $R = 1$ and $A = a$, but with low corresponding predicted probabilities, can have disproportionate influence on the final estimate of ψ_a

- Major source of **instability** (Chattopadhyay et al. 2024)

Equivalence of Approach 1 and Approach 2 EICs

$$\phi_a^{\mathsf{obs}}(\boldsymbol{O}) = \frac{R}{\mathbb{P}(R=1|\boldsymbol{Z})}\chi_a^{\mathsf{full}}(\boldsymbol{O}) - \left(\frac{R}{\mathbb{P}(R=1|\boldsymbol{Z})} - 1\right)\varphi_a(\boldsymbol{Z})$$

Equivalence of Approach 1 and Approach 2 EICs

$$\phi_a^{\mathsf{obs}}(\boldsymbol{O}) = \frac{R}{\mathbb{P}(R=1|\boldsymbol{Z})}\left[\frac{I(A=a)}{g_a(\boldsymbol{X})}(Y - m_a(\boldsymbol{X})) + m_a(\boldsymbol{X}) - \psi_a\right] - \left(\frac{R}{\mathbb{P}(R=1|\boldsymbol{Z})} - 1\right)\varphi_a(\boldsymbol{Z})$$

Equivalence of Approach 1 and Approach 2 EICs

$$\phi_a^{\mathsf{obs}}(\boldsymbol{O}) = \frac{R}{\mathbb{P}(R=1|\boldsymbol{Z})}\left[\frac{I(A=a)}{\pi_a(\boldsymbol{X})}\left(Y - \frac{\eta_a(\boldsymbol{X})}{\pi_a(\boldsymbol{X})}\right) + \frac{\eta_a(\boldsymbol{X})}{\pi_a(\boldsymbol{X})} - \psi_a\right]$$
$$- \left(\frac{R}{\mathbb{P}(R=1|\boldsymbol{Z})} - 1\right)\varphi_a(\boldsymbol{Z})$$

Equivalence of Approach 1 and Approach 2 EICs

$$\phi_a^{\mathsf{obs}}(\boldsymbol{O}) = \frac{R}{\mathbb{P}(R=1|\boldsymbol{Z})}\left[\frac{I(A=a)}{\pi_a(\boldsymbol{X})}\left(Y - \frac{\eta_a(\boldsymbol{X})}{\pi_a(\boldsymbol{X})}\right) + \frac{\eta_a(\boldsymbol{X})}{\pi_a(\boldsymbol{X})} - \psi_a\right]$$
$$- \left(\frac{R}{\mathbb{P}(R=1|\boldsymbol{Z})} - 1\right)\left(\frac{\lambda_a(\boldsymbol{Z})\mu_a(\boldsymbol{Z})}{\pi_a(\boldsymbol{X})} - \frac{\lambda_a(\boldsymbol{Z})\eta_a(\boldsymbol{X})}{\pi_a(\boldsymbol{X})^2} + \frac{\eta_a(\boldsymbol{X})}{\pi_a(\boldsymbol{X})} - \psi_a\right)$$

Approach 1 EIF

Following Approach 1, the efficient influence curve of ψ_a can be written

$$\phi_a(\boldsymbol{O}) = \frac{\eta_a(\boldsymbol{X})}{\pi_a(\boldsymbol{X})} - \psi_a + \frac{\lambda_a(\boldsymbol{Z})}{\pi_a(\boldsymbol{X})}\left(\mu_a(\boldsymbol{Z}) - \frac{\eta_a(\boldsymbol{X})}{\pi_a(\boldsymbol{X})}\right) + \frac{RI(A=a)}{\kappa(\boldsymbol{Z})\pi_a(\boldsymbol{X})}\left(Y - \frac{\eta_a(\boldsymbol{X})}{\pi_a(\boldsymbol{X})}\right) - \frac{R\lambda_a(\boldsymbol{Z})}{\kappa(\boldsymbol{Z})\pi_a(\boldsymbol{X})}\left(\mu_a(\boldsymbol{Z}) - \frac{\eta_a(\boldsymbol{X})}{\pi_a(\boldsymbol{X})}\right) . \tag{1}$$

where $\boldsymbol{O} = (Y, A, R, \boldsymbol{Z})$ and $\boldsymbol{Z} = (\boldsymbol{X}, A^*, Y^*)$

Approach 2 EIF

Let

$$\chi_a(\boldsymbol{O}) = \frac{I(A=a)}{g_a(\boldsymbol{X})}(Y - m_A(\boldsymbol{X})) + m_a(\boldsymbol{X}) - \psi_a$$

denote the EIF for ψ_a in a complete data model. Then, the EIF for ψ_a under the observed (two-phase sampling) model is

$$\phi_a(\boldsymbol{O}) = \frac{R\chi_a(\boldsymbol{O})}{\kappa(\boldsymbol{Z})} - \left(\frac{R}{\kappa(\boldsymbol{Z})} - 1\right)\varphi_a(\boldsymbol{Z})$$

where $\varphi_a(\boldsymbol{Z}) = \mathbb{E}[\chi_a(\boldsymbol{O})|\boldsymbol{Z}, R=1]$

Approach 1: Rate Robustness

Recall that $\lambda_a(\boldsymbol{Z}) = \mathbb{P}(A = a|\boldsymbol{Z}, R = 1)$ and $\mu_a(\boldsymbol{Z}) = \mathbb{E}[Y|A = a, \boldsymbol{Z}, R = 1]$ and define

$$\pi_a(\boldsymbol{X}) = \mathbb{E}[\lambda_a(\boldsymbol{Z})|\boldsymbol{X}] \quad \text{and} \quad \eta_a(\boldsymbol{X}) = \mathbb{E}[\lambda_a(\boldsymbol{Z}) \cdot \mu_a(\boldsymbol{Z})|\boldsymbol{X}]$$

Further let $\kappa(\boldsymbol{Z}) = \mathbb{P}(R = 1|\boldsymbol{Z})$. Then, it can be shown

$$\begin{aligned}\mathbb{E}[\hat{\psi}_a^{\mathsf{OS},1} - \psi_a] = O_{\mathbb{P}}\Bigg(\frac{1}{\sqrt{n}} \quad &+ \\ ||\hat{\pi}_a - \pi_a|| \cdot \{||\hat{\pi}_a - \pi_a|| + ||\hat{\eta}_a - \eta_a||\} \quad &+ \\ ||\hat{\kappa} - \kappa|| \cdot \{||\hat{\lambda}_a - \lambda_a|| + ||\hat{\mu}_a - \mu_a||\}\Bigg)\end{aligned}$$

Approach 1: ~~Double~~ Awkward Robustness

Recall that $\lambda_a(\boldsymbol{Z}) = \mathbb{P}(A = a|\boldsymbol{Z}, R = 1)$ and $\mu_a(\boldsymbol{Z}) = \mathbb{E}[Y|A = a, \boldsymbol{Z}, R = 1]$ and define

$$\pi_a(\boldsymbol{X}) = \mathbb{E}[\lambda_a(\boldsymbol{Z})|\boldsymbol{X}] \quad \text{and} \quad \eta_a(\boldsymbol{X}) = \mathbb{E}[\lambda_a(\boldsymbol{Z}) \cdot \mu_a(\boldsymbol{Z})|\boldsymbol{X}]$$

Further let $\kappa(\boldsymbol{Z}) = \mathbb{P}(R = 1|\boldsymbol{Z})$. Then, it can be shown $\hat{\psi}_a^{\text{OS,1}} \xrightarrow{p} \psi_a$ if

1. $||\hat{\kappa} - \kappa|| = o_{\mathbb{P}}(1)$ OR both $||\hat{\mu}_a - \mu_a|| = o_{\mathbb{P}}(1)$ and $||\hat{\lambda}_a - \lambda_a|| = o_{\mathbb{P}}(1)$

2. $||\hat{\pi}_a - \pi_a|| = o_{\mathbb{P}}(1)$

Interestingly, consistency of $\hat{\psi}_a^{\text{OS,1}}$ <u>requires</u> consistent estimation of π_a (which will be heavily dependent on $\hat{\lambda}_a$)

- Similar requirements for $\sqrt{n}$ consistency + asymptotic normality

Approach 2: Double Robustness

It can be shown $\hat{\psi}_a^{\text{OS,2}} \xrightarrow{p} \psi_a$ if

1. Either $||\hat{m}_a - m_a|| = o_{\mathbb{P}}(1)$ or $||\hat{g}_a - g_a|| = o_{\mathbb{P}}(1)$, and
2. Either $||\hat{\kappa} - \kappa|| = o_{\mathbb{P}}(1)$ or $||\hat{g}_a - g_a|| = o_{\mathbb{P}}(1)$,

Similar requirements for $\sqrt{n}$ consistency + asymptotic normality

In settings where κ is known (common in two-phase designs) notice above requirements collapse to standard double-robustness condition

Anchor and Augmentation Estimation (Han et al., 2025)

Consider a setting where one

- Is concerned $\hat{\psi}_a^{\mathsf{OS},1}$ is inconsistent due to model misspecification, but
- Is confident $\hat{\psi}_a^{\mathsf{OS},2}$ is consistent

Can consider estimators of the form

$$\hat{\psi}_a(b) = \hat{\psi}_a^{\mathsf{OS},2} - b(\hat{\psi}_a^{\mathsf{OS},2} - \hat{\psi}_a^{\mathsf{OS},1})$$

where one chooses b to minimize

$$L(b) = \widehat{\mathsf{Var}}(\hat{\psi}_a(b)) + \lambda \cdot (\hat{\psi}_a^{\mathsf{OS},1} - \hat{\psi}_a^{\mathsf{OS},2})^2 \cdot |b|$$

Can be viewed as a penalized control variates estimator (Yang and Ding, 2019) and has been used to data-adaptively construct aggregated estimators in federated settings (Han et al., 2025)

Approach 2: Empirical efficiency maximization

Recall Approach 2's one-step estimator and bias structure

$$\hat{\psi}_a^{\text{OS,2}} = \hat{\psi}_a^{\text{PI,2}} + \frac{1}{n}\sum_{i=1}^{n}\left[\frac{R_i}{\mathbb{P}(R_i = 1|\boldsymbol{Z}_i)}\hat{\chi}_a^{\text{full}}(\boldsymbol{O}_i; \hat{m}_a, \hat{g}_a) - \left(\frac{R_i}{\mathbb{P}(R_i = 1|\boldsymbol{Z}_i)} - 1\right)\hat{\varphi}_a(\boldsymbol{Z}_i)\right]$$

$$\hat{\psi}_a^{\text{OS,2}} - \psi_a = O_{\mathbb{P}}\left(\frac{1}{\sqrt{n}} + ||\hat{m}_a - m_a|| \cdot ||\hat{g}_a - g_a|| + ||\hat{\kappa} - \kappa|| \cdot ||\hat{\varphi}_a - \varphi_a||\right)$$

Given this bias structure, and recalling $\kappa(\boldsymbol{Z}) = \mathbb{P}(R = 1|\boldsymbol{Z})$ will often be known, why not obtain $\hat{\varphi}_a$ in a way that minimizes the empirical variance of $\hat{\psi}_a^{\text{OS-ALT}}$?

- This notion has been proposed before in the context of clinical trials (Rubin and van der Laan 2008)

Empirical efficiency maximization (EEM)

High level: Rather than choose $\hat{\varphi}_a$ aiming to minimize $||\hat{\varphi}_a - \varphi_a||$, choose $\hat{\varphi}_a$ to minimize

$$\frac{1}{n}\sum_{i=1}^{n}\left[\frac{R_i}{\mathbb{P}(R_i = 1|\boldsymbol{Z}_i)}\hat{\chi}_a^{\text{full}}(\boldsymbol{O}_i; \hat{m}_a, \hat{g}_a) - \left(\frac{R_i}{\mathbb{P}(R_i = 1|\boldsymbol{Z}_i)} - 1\right)\hat{\varphi}_a(\boldsymbol{Z}_i)\right]^2$$

Key idea: Asymptotically (and under correct model specification) the true $\varphi_a(\boldsymbol{Z})$ will minimize the variance of $\hat{\psi}_a^{\text{OS},2}$

- Can target this variance-minimizing property directly

With a minor re-parameterization, the above minimization problem can be carried out with existing regression software

Empirical efficiency maximization (EEM)

Regress the transformed pseudo-outcome

$$\left(\frac{R_i}{\mathbb{P}(R_i = 1|\boldsymbol{Z}_i)} - 1\right)^{-1} \frac{R_i}{\mathbb{P}(R_i = 1|\boldsymbol{Z}_i)} \hat{\chi}_a^{\text{full}}(\boldsymbol{O}_i; \hat{m}_a, \hat{g}_a)$$

on $\boldsymbol{Z}$, adding regression weights of the form

$$w_a(\boldsymbol{O}) = \left(\frac{R_i}{\mathbb{P}(R_i = 1|\boldsymbol{Z}_i)} - 1\right)^2$$

Intuition: Small propensity scores will blow up weights $w_a(\boldsymbol{O})$ – will be further amplified if the residual $\hat{\chi}_a^{\text{full}}(\boldsymbol{O}_i; \hat{m}_a, \hat{g}_a) - \hat{\varphi}_a(\boldsymbol{Z})$ is large

- Use of weights $w_a(\boldsymbol{O})$ "prioritize" minimizing residuals for units with large inverse weights